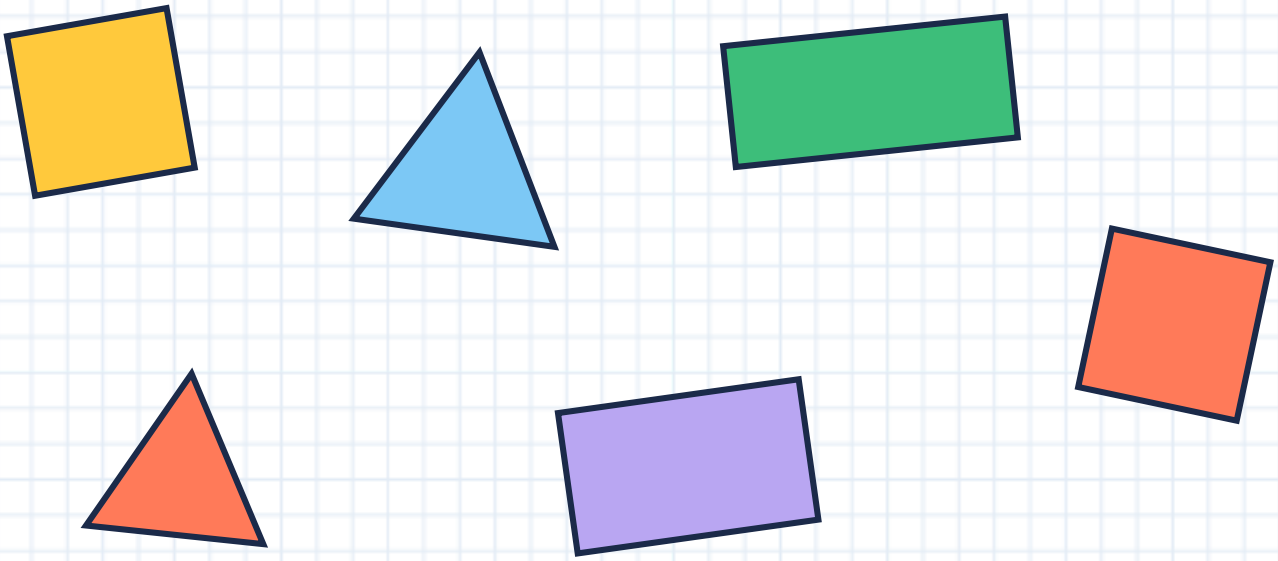


LEVEL 5 · HOLIDAY PROGRAM

Math Inquirers

Notice it. Test it. Explain why.

Name _____ **Group** _____

My Inquirer Rules

1. Notice the pattern.
2. Test it with objects first.
3. Check your answer.
4. Explain why.

My Inquiry Journal

Your tutor stamps each lesson.

Lesson 1
Million Puzzles

Lesson 2
Fraction Feats

Lesson 3
Before and After

Lesson 4
Triangle Areas

Lesson 5
Volume Vault

Lesson 6
Rate Racers

Lesson 7
Percentage Puzzles

Lesson 8
Angle Chasers

Lesson 9
Pattern Generalisers

Lesson 10
Challenge Day

Tutor: _____ Program dates: _____

MILLION PUZZLES

Name _____ Group _____ Date _____

Start here

After question 2: carry on, or try Think harder. Swap any time.

No.1

Place value. Read big numbers in groups of three from the right: 4 | 053 | 200.

a Numerals for five million, three hundred and eight thousand and forty: _____

b In 6 482 315, the digit 4 stands for _____. The ten thousands digit is _____.

c How many thousands make 3 250 000? _____

No.2

Use ALL seven digit cards, each once, to make a 7-digit number (it cannot start with 0).

0	2	3	5	6	8	9

a Greatest: _____ (b) Smallest EVEN: _____

c Closest to 6 000 000: _____ Test both sides of the target!

Best just below: _____ It is _____ away.

Best just above: _____ It is _____ away.

No.3

Digit riddle (try this one if you like). Use the clues one at a time, then check them all.

Clue 1. I am a 7-digit number. My digits are all different.
Clue 2. Rounded to the nearest hundred thousand, I am 7 300 000.
Clue 3. My ten thousands digit is 3 more than my hundred thousands digit.
Clue 4. My thousands digit = my ten thousands digit – my hundred thousands digit.
Clue 5. My hundreds digit is twice my thousands digit. My tens digit is 0.
Clue 6. My digits add up to 32.
START HERE: I start 7 2 ... or 7 3 ... Which start fits Clue 3?

M	HTh	TTh	Th	H	T	O

I am _____

Start here: page 2

Page 2. Start with Q4 and Q5 (a) (b), then choose. You do not have to finish. Last 10 minutes: Think harder, Go further or the Investigation.

No.4

Ravi used a calculator. Estimate each answer by rounding. Which answers must be wrong?

Calculation	Ravi's answer	My estimate	Could be right / Must be wrong
(a) 4987 × 21	104 727		
(b) 3 004 512 ÷ 48	6259		
(c) 2 345 678 + 3 987 654	5 333 332		
(d) 806 × 395	318 370		

No.5

Brackets first, then × and ÷, then + and –. Put in ONE pair of brackets to make each statement true.

a 6 + 4 × 5 – 2 = 48

b 6 + 4 × 5 – 2 = 18

c 1 000 000 ÷ 200 + 300 × 2 = 4000

d Target 100! Use 3, 5, 7 and 9 once each with +, –, ×, ÷ and brackets to make 100. Hint: 100 = 5 × 20.

No.6

A 7-digit number rounded to the nearest thousand is 2 500 000. Its digits add up to 9.

Which numbers round to 2 500 000?



Start here: numbers from 2 500 000 up begin 2 5 0 … Those three digits add to 7, so the last four digits must add to _____. Try 2 500 002 first.

a Write the range first. The smallest number that rounds to 2 500 000 is _____ and the greatest is _____.

b Find ALL the numbers. Make a list in order. How do you know you have them all?

INVESTIGATION

Name _____ Group _____ Date _____

Investigation: What if we change the target?

An open puzzle for everyone. Easy first step: use the cards 0, 1, 2, 4, 5, 7, 8 for just ONE target, 4 000 000. Write one number from just below it and one from just above it. Which is closer? Then do part (a).

No.8

(a) Use the digit cards 0, 1, 2, 4, 5, 7, 8, each once, to make the 7-digit number closest to a TARGET. Find the closest number for the targets 3 000 000, 4 000 000, 5 000 000 and 6 000 000.

(b) For each target, does the winner come from just BELOW the target or just ABOVE it? What do you notice? Try 7 000 000 and 8 000 000 to test your idea.

(c) What stays the same in every case? Write a rule that tells someone which side wins.

My reflection (2 minutes. Your tutor may read this to plan the next lesson. It is never marked or shown to others. "Pass" is always OK.)

Something I did well

_____

The hardest part was

_____

What helped me: ☐ a picture or model ☐ a list or table ☐ a simpler number ☐ my partner ☐ I asked for help ☐ checking my estimate

Next time I will try

_____

☐ I am still stuck (we will look at it together next lesson, in class) ☐ Pass

Meet a mathematician: Dorothy Vaughan (1910–2008, United States. Optional.)

Dorothy Vaughan was a maths teacher who joined the NACA laboratory at Langley (it later became NASA) in 1943. She became head of a team of "human computers" who did long calculations by hand. When electronic computers arrived, she taught herself the FORTRAN programming language and taught it to her team too.

Try it (optional): A human computer checks two big numbers: 2 450 000 and 1 375 000. Work out their sum and their difference. Then work out 1250 × 400.

My answer _____

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Think harder

No.1

How many 7-digit numbers have digits that add up to 2? List them all.

No.2

Put ONE pair of brackets into 2 + 3 × 4 + 5 to make 25. Then put one pair in a different place to make 29.

No.3

A whole number rounded to the nearest thousand is 640 000. What is the greatest it could be? The smallest?

No.4

Use each of the digits 1, 2, 3, 4, 5, 6 and 7 once. What is the difference between the greatest and the smallest 7-digit number you can make?

Go further (a puzzle to chew on)

No.5

Put brackets into 1 + 2 × 3 + 4 × 5 + 6 to make the answer as big as possible. You may use more than one pair. What is the biggest answer?

No.6

I am a 7-digit odd number. My digits are all different and add up to 42. Rounded to the nearest hundred thousand, I am 7 000 000. I am the smallest number that fits. Who am I?

No.7

Ali writes all the whole numbers from 1 to 1000. How many digits does he write? (Go further: how many if he writes all the numbers from 1 to 1 000 000?)

FRACTION FEATS

Name _____ Group _____ Date _____

Start here

After question 3: carry on, or try Think harder. Swap any time.

No.1

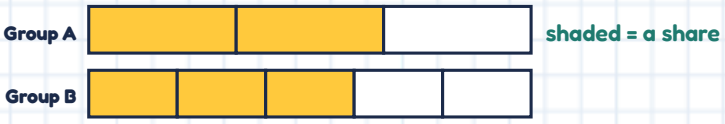
Sharing. Remember: a ÷ b = a/b.

a 5 pizzas are shared equally among 8 children. Each gets _____ of a pizza.

b A 9 m ribbon is cut into 4 equal pieces. Each piece is _____ m long. (a fraction or mixed number)

c Group A shares 2 pizzas among 3 children. Group B shares 3 pizzas among 5 children. In which group does each child get more? Explain.

Hint: how much of a whole pizza is each child short of?



No.2

Write each fraction as the sum of two DIFFERENT unit fractions (1 on top). Take away 1/2, then 1/3 ... Is what is left a unit fraction?

Start here: $\frac{7}{10} - \frac{1}{2}$. Write $\frac{1}{2}$ as tenths: $\frac{1}{2} = \frac{__}{10}$. So $\frac{7}{10} - \frac{1}{2} = \frac{__}{10} = \frac{1}{____}$.

a $\frac{7}{10} = \frac{1}{\square} + \frac{1}{\square}$ **(b)** $\frac{9}{20} = \frac{1}{\square} + \frac{1}{\square}$ (hint: $\frac{1}{4} = \frac{__}{20}$)

c Find TWO ways for $\frac{3}{8}$: $\frac{3}{8} = \frac{1}{\square} + \frac{1}{\square}$ and $\frac{3}{8} = \frac{1}{\square} + \frac{1}{\square}$

Test 1/3, 1/4, 1/5 ... in order. How do you know when to stop?

No.3

Siti had 240 stickers. She gave $\frac{1}{4}$ of them to Ben and $\frac{2}{3}$ of the REMAINDER to Ali.



a How many stickers did Ali get? _____

b How many stickers did Siti have left? _____

c What fraction of her 240 stickers did Siti have left? _____

Start here: page 2

Page 2. Start with Q5 and Q4 (a) (b), then choose. You do not have to finish. Last 10 minutes: Think harder, Go further or the Investigation.

No.4

Look for a pattern: $\frac{1}{2}$, $\frac{1}{6}$, $\frac{1}{12}$, $\frac{1}{20}$, ...

a Write the next two fractions: _____ , _____ What do you notice about the bottom numbers?

b Work out each sum. Give each answer in its simplest form.

$\frac{1}{2} + \frac{1}{6} =$	
$\frac{1}{2} + \frac{1}{6} + \frac{1}{12} =$	
$\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{20} =$	

c Predict: $\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \ldots + \frac{1}{90} =$ _____ Explain how you know.

No.5

Ravi spent $\frac{2}{5}$ of his money on a game and $\frac{1}{2}$ of the remainder on a book. He had $36 left. Draw a model.

Start here: draw 5 equal parts; the game takes 2. Cut the rest to find half of it.

a How much money did Ravi have at first? _____

b How much did the book cost? _____

No.6

Use the digit cards 1 2 3 4 **once each in the boxes (fractions need not be proper).**

$$\frac{\square}{\square} + \frac{\square}{\square}$$

a The largest sum: _____

b The smallest sum: _____ Did you test both ways?

c The sum closest to 1: _____

INVESTIGATION

Name _____ Group _____ Date _____

Investigation: Is it always true? Test and explain

An open puzzle for everyone. Easy first step: cut a bar into 6 equal parts. 1/2 is ____ parts and 1/3 is ____ parts, so 1/2 – 1/3 = ____/6. Now try part (a).

No.8

(a) Take two unit fractions that are next to each other, such as 1/3 and 1/4. Take the smaller away from the bigger: 1/3 – 1/4 = _____. Try at least four more pairs (1/4 and 1/5, 1/5 and 1/6, 1/9 and 1/10, 1/20 and 1/21).

(b) What do you notice about the answer each time? Write your idea in words. Is it always true? Explain with a drawing (hint: cut a bar into 3 × 4 = 12 equal parts).

(c) Use your idea to write 1/6 + 1/12 + 1/20 as one fraction without finding a common denominator. Then make up your own sum of this kind (three or four fractions) for a partner to solve.

My reflection (2 minutes. Your tutor may read this to plan the next lesson. It is never marked or shown to others. "Pass" is always OK.)

Something I did well

_____

The hardest part was

_____

What helped me: ☐ a picture or model ☐ a list or table ☐ a simpler number ☐ my partner ☐ I asked for help ☐ checking my estimate

Next time I will try

_____

☐ I am still stuck (we will look at it together next lesson, in class) ☐ Pass

Meet a mathematician: Eugenia Cheng (born 1976, England and the United States. Optional.)

Eugenia Cheng is a mathematician, born in England in 1976, who explains maths with food. Her book How to Bake Pi (2015) opens each chapter with a dessert recipe, and one example is about three brothers and sisters trying to share a cake.

Try it (optional): Three children share a cake. Ana eats 1/3 of it and Ben eats 1/4 of it. What fraction of the cake is left for the third child?

My answer _____

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Think harder

No.1

Can 4/5 be written as the sum of two DIFFERENT unit fractions? Test in order and explain how you know when to stop.

No.2

Ben ate 1/3 of a pizza. Ali ate 1/4 of what was left. What fraction of the pizza is left?

No.3

Siti shares 3 cakes equally among 8 children. Ben shares 2 cakes equally among 5 children. Whose child gets more cake, and how much more?

No.4

Mei walked 3/5 of a path in the morning and 1/4 of the WHOLE path in the afternoon. She still had 600 m to walk. How long is the path?

Go further (a puzzle to chew on)

No.5

Work out 1/6 + 1/12 + 1/20 + 1/30 + 1/42 without finding a common denominator for all five. (Hint: 1/6 = 1/2 – 1/3.)

No.6

5 pizzas are shared equally among 12 children. Each child must get exactly 2 pieces, and each piece must be a unit fraction of a pizza (like 1/3 or 1/4). How can the pizzas be cut?

No.7

A tank is full of water. Every hour, 1/3 of the water that is in the tank is let out. After 3 hours there are 16 litres left. (a) How much water was in the full tank? (b) After how many hours is less than half of the original water left? (c) How much water has been let out by then?

BEFORE AND AFTER

Name _____ Group _____ Date _____

Start here

After question 3: carry on, or try Think harder. Swap any time.

Draw a model. Ask: what stays the same?

No.1

Ravi had some money. He spent 1/3 of it on a book and 1/4 of the remainder on a pen. He had $36 left.

Ravi's money

Divide the bar: the book, the pen and what is left.

a Try $60 first. The book costs $_____, the pen costs $_____, and $_____ is left.

b Ravi really had $36 left, not $30. Was $60 too much or too little? How much money did Ravi have at first?

Answer: _____

c How much more did the book cost than the pen?

Answer: _____

d (try this one if you like) What fraction of his money did he spend on the pen? _____

No.2

Siti and Mei had 84 stickers altogether. Siti gave Mei 12 stickers. Then Siti had 3/4 as many stickers as Mei.

After Siti gave Mei 12 stickers:

Siti 

Mei 

Draw the Before model on the right of this.

a How many stickers altogether after the swap? _____ The 7 boxes hold this many, so one box = _____ stickers.

b How many stickers did Siti have at first? How many did Mei have at first?

Siti: _____ **Mei:** _____

Start here: page 2

Page 2. Start with Q3 (a), Q4 (a) (b) and Q5 (a) (b), then choose. You do not have to finish. Last 10 minutes: Think harder, Go further or the Investigation.

No.3

Ali has 50 marbles and Ben has 30 marbles. Ali gives some of his marbles to Ben.

a Ben says Ali should give 20 marbles so they have the same number. Test it: after Ali gives 20, Ali has _____ and Ben has _____. Is Ben right? Explain.

b The total stays _____. After: Ali 3 parts, Ben 5 parts, so one part = _____. How many marbles must Ali give Ben so that Ali has 3/5 as many as Ben?

Answer: _____

No.4

Ravi had $50 and Ali had $26. Their mother gave each of them the same amount of money. Then Ali had 2/3 as much money as Ravi.

Start here: try a gift of $10 each. Is Ali's money then 2/3 of Ravi's? Is the gift too big or too small?

a What stays the same: the total or the difference? How much is it?

Answer: _____

b How much money did their mother give each boy?

Answer: _____

No.5

A box had red beads and blue beads. 3/7 of the beads were red. Then 30 red beads were added. Now 3/5 of the beads are red.

a Which colour of beads did not change?

Answer: _____

b Before: red 3 units, blue _____ units. After: 5 equal parts, 3 red and 2 blue (draw them below). Blue is still _____ units, so 2 parts = _____ units and the whole box = _____ units.

The box after



Shade 3 parts red and 2 parts blue.

c How many beads were in the box at first?

Answer: _____

d How many beads are in the box now?

Answer: _____

INVESTIGATION

Name _____ Group _____ Date _____

Investigation: What if we change the total?

An open puzzle for everyone. Easy first step: the total is 48 + 12 = 60. Draw one bar for 60 and cut it into 3 equal parts. How many stickers is one part? Then do part (a).

No.8

(a) Ali has 48 stickers and Mei has 12. Ali gives Mei some stickers. Afterwards Ali has exactly twice as many as Mei. How many does Ali give? Draw the After bars.

(b) What if the total is different? Try the totals 45, 50, 30 and 60 in any way you like (Ali has more than Mei to start). For which totals can Ali end up with exactly TWICE as many as Mei, with whole stickers? For which can it NOT be done?

(c) What stays the same? Write a rule that tells you which totals work, and explain with a bar.

My reflection (2 minutes. Your tutor may read this to plan the next lesson. It is never marked or shown to others. "Pass" is always OK.)

Something I did well

_____

The hardest part was

_____

What helped me: ☐ a picture or model ☐ a list or table ☐ a simpler number ☐ my partner ☐ I asked for help ☐ checking my estimate

Next time I will try

_____

☐ I am still stuck (we will look at it together next lesson, in class) ☐ Pass

Meet a mathematician: Alcuin of York (about 735–804, England and Charlemagne's court. Optional.)

Alcuin of York was a monk and teacher from Northumbria who helped the emperor Charlemagne run schools. A Latin book of about fifty number puzzles, Problems to Sharpen the Young, is linked with him. Historians think he collected the puzzles, so he may not have invented them all.

Try it (optional): A puzzle in the style of Alcuin's stork puzzle (our own wording): 3 times the number of storks, plus half the number of storks, plus 2 makes 100. How many storks are there? Hint: draw 3 and a half units.

My answer _____

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Think harder

No.1

Ravi spent 1/4 of his money on Monday and 2/5 of the remainder on Tuesday. He spent $66 altogether. How much money did he have at first?

No.2

A box had red pens and blue pens. 2/3 of the pens were blue. After 12 blue pens were taken out, 1/2 of the pens were blue. How many pens were in the box at first?

No.3

Mrs Tan is 41 years old and her son is 9. How many years ago was her son's age 1/5 of her age?

No.4

Ben and Siti had $140 altogether. Ben spent 1/2 of his money and Siti spent $20. Then they had the same amount left. How much did Ben have at first?

Go further (a puzzle to chew on)

No.5

Mei had $200 and Siti had $80. Mei spent 3 times as much money as Siti. Then they had the same amount left. How much money did Siti spend?

No.6

Ravi had some stickers. He gave 1/2 of them and 4 more to Ali. Then he gave 1/2 of the stickers he had left and 4 more to Ben. He had 4 stickers left. How many stickers did he have at first?

No.7

Ali and Ben had 140 stamps altogether. Ali gave away 1/3 of his stamps and Ben gave away 1/2 of his. Then they had the same number of stamps left. How many stamps did Ben give away?

TRIANGLE AREAS

Name _____ Group _____ Date _____

Start here

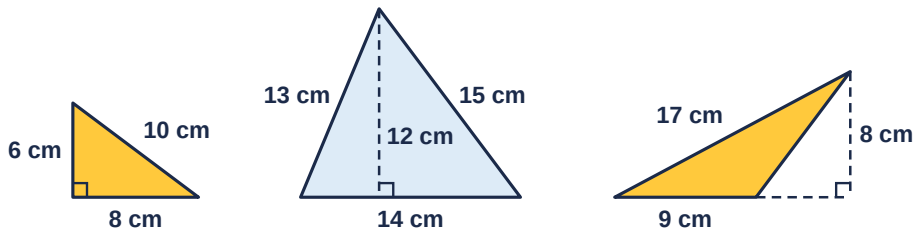
After question 3: carry on, or try Think harder. Swap any time.

Area of a triangle = 1/2 × base × height. The height meets the base at a right angle.

No.1

Find the area of each triangle. The drawings are to scale.

(a) (b) (c)



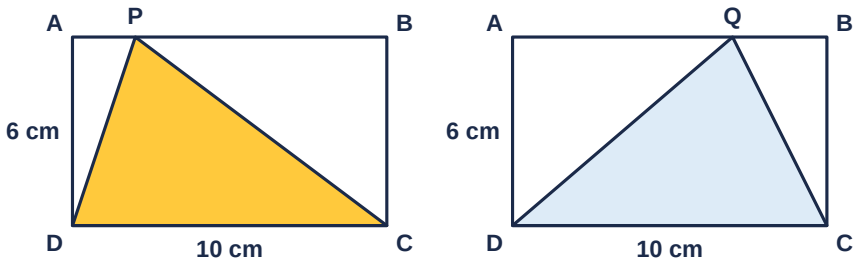
Start here: the little square marks a right angle. The height makes a right angle with the base; a slanting side is NOT the height. Dashed lines show a height. In (c) it is outside the triangle.

a Area of (a) = _____ cm² (b) Area of (b) = _____ cm² (c) Area of (c) = _____ cm²

d In (b), which lengths did you NOT need? _____

No.2

ABCD is a rectangle 10 cm by 6 cm. P and Q are points on AB.



a Area of triangle PDC = _____ cm² (b) Area of triangle QDC = _____ cm²

c Explain why the two triangles have the same area.

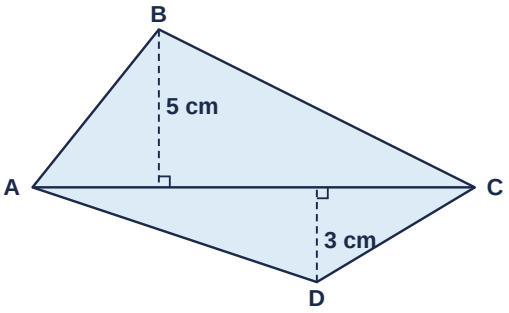
.....

Start here: page 2

Page 2. Start with Q5 and Q3, then choose. You do not have to finish. Last 10 minutes: Think harder, Go further or the Investigation.

No.3

Find the area of the figure ABCD. Split it into two triangles along AC.



AC = 14 cm

Start here: AC = 14 cm is the base of BOTH triangles. Read the two heights from the picture.

a Height of triangle ABC = _____ cm. Area of ABC = _____ cm^2

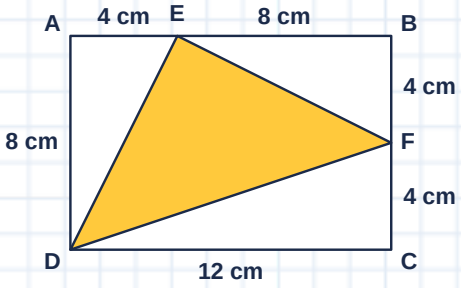
b Height of triangle ACD = _____ cm. Area of ACD = _____ cm^2

c Area of ABCD = _____ cm^2

No.4

ABCD is a rectangle. Find the area of the shaded triangle DEF.

Start here: find the three white triangles ADE, EBF and FCD, then take them away from the rectangle.



a Area of the rectangle ABCD = _____ cm^2

b ADE = _____ cm^2 EBF = _____ cm^2 FCD = _____ cm^2

c Area of the shaded triangle DEF = _____ cm^2

No.5

Mei draws a triangle with a base of 8 cm and a height of 6 cm.

a What is the area of Mei's triangle? _____ cm^2

b Siti's triangle has the same area but a base of 12 cm. What is its height? _____ cm

c One more triangle with the same area: base _____ cm, height _____ cm

INVESTIGATION

Name _____ Group _____ Date _____

Investigation: Find all the ways

An open puzzle for everyone. Easy first step: draw a right-angled triangle on squared paper with short sides 3 cm and 4 cm. What is its area? Then look for others with area 6 cm².

No.8

(a) On square grid paper (1 square = 1 cm), draw right-angled triangles whose two short sides are along the grid lines and are whole numbers of cm, and whose area is exactly 6 cm². Find ALL of them (a triangle turned round is the same triangle). How do you know you have them all?

(b) Now keep the base 4 cm along the bottom. Mark where the top corner can go so that the area is still 6 cm². What do you notice about all the possible top corners?

(c) What if the area is 10 cm²? 12 cm²? How many right-angled triangles can you find each time? Which area gave the most?

My reflection (2 minutes. Your tutor may read this to plan the next lesson. It is never marked or shown to others. "Pass" is always OK.)

Something I did well

_____

The hardest part was

_____

What helped me: ☐ a picture or model ☐ a list or table ☐ a simpler number ☐ my partner ☐ I asked for help ☐ checking my estimate

Next time I will try

_____

☐ I am still stuck (we will look at it together next lesson, in class) ☐ Pass

Meet a mathematician: Liu Hui (3rd century CE, China (kingdom of Wei). Optional.)

Liu Hui lived in China in the 3rd century. In the year 263 he wrote a commentary on The Nine Chapters on the Mathematical Art, explaining why its rules for the areas of fields work. He often cut shapes into pieces and moved the pieces around.

Try it (optional): A right-angled triangle has two short sides of 6 cm and 8 cm. Two copies fit together to make a 6 cm by 8 cm rectangle. What is the area of one triangle?

My answer _____

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know. Draw a big, neat diagram first and label every length or angle you know.

Think harder

No.1

The area of a right-angled triangle is 24 cm². Its two short sides (the ones that make the right angle) differ by 2 cm. How long are the two short sides?

No.2

A right-angled triangle has sides of 12 cm, 16 cm and 20 cm. What is its area?

No.3

A triangle and a square have the same area. The square has sides of 6 cm. The triangle has a base of 9 cm. What is the height of the triangle?

No.4

A triangle and a rectangle have the same base, 12 cm, and the same height, 7 cm. What is the difference between their areas?

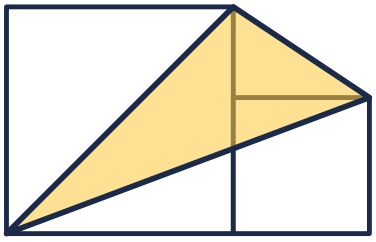
Go further (a puzzle to chew on)

No.5

A right-angled triangle has sides of 15 cm, 20 cm and 25 cm. What is its height when the 25 cm side is the base?

No.6

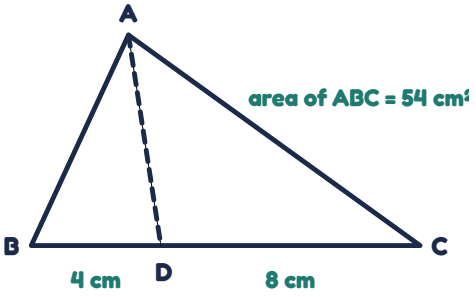
Two squares stand side by side on a line, touching: a big square of side 10 cm and a small square of side 6 cm. Join the bottom-left corner of the big square, the top-right corner of the big square and the top-right corner of the small square to make a triangle. What is its area? Try again with a small square of side 4 cm. What do you notice?



big square: 10 cm 6 cm

No.7

Triangle ABC has an area of 54 cm². D is a point on BC with BD = 4 cm and DC = 8 cm. What is the area of triangle ABD?



VOLUME VAULT

Name _____ Group _____ Date _____

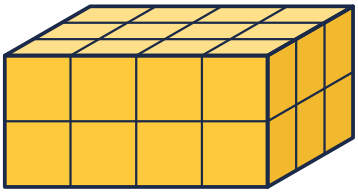
Start here

After question 3: carry on, or try Think harder. Swap any time.

Words and signs. cm^3 = cubic centimetres (one small cube). 1 ℓ = 1000 cm^3. Breadth means how wide.

No.1

This solid cuboid is built from 1-cm cubes.



(a) What is its volume? ..... cm^3

(b) Mei adds 1-cm cubes (she takes none away) until the solid is a cube. What is the fewest cubes she must add? Explain.

.....

No.2

Ravi has 36 one-centimetre cubes. He uses all of them to build one cuboid. Turning a cuboid round does not make a new one.

Start here: 1×1×36, 1×2×18, 1×3×____, 1×4×____, 1×6×____; then shortest edge 2, then 3.

a List every cuboid he can build (length × breadth × height). Work in order. How many are there?

.....

.....

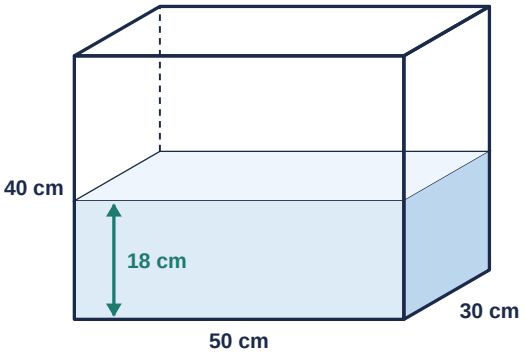
b (try this one if you like) Which of them can fit inside a box 5 cm long, 5 cm wide and 5 cm high?

_____

No.3

A glass tank is 50 cm long, 30 cm wide and 40 cm high. The water in it is 18 cm deep.

Start here: 18 layers of cubes. One layer is 50 × 30 = _____ cubes, so 18 layers = _____ cm^3. 1000 cm^3 = 1 ℓ. A calculator is fine.



(a) How many litres of water are in the tank?

..... ℓ

(b) How many more litres are needed to fill it to the top?

..... ℓ

(c) Instead, 9 ℓ of water is let out. How deep is the water now?

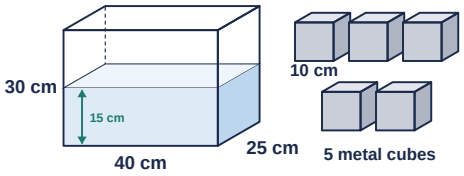
..... cm

Start here: page 2

Page 2. Start with Q4 (a) to (c) and Q5, then choose. You do not have to finish. Last 10 minutes: Think harder, Go further or the Investigation.

No.4

A tank has a base 40 cm by 25 cm. The water in it is 15 cm deep. Ali drops in 5 metal cubes with edges of 10 cm. They sink to the bottom and are completely under water.

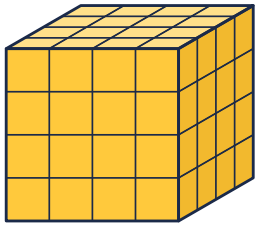


a Volume of 5 cubes: _____ cm³ (b) Rise: _____ cm (c) Depth now: _____ cm

d Then Ali drops in a stone. The water rises another 3 cm. What is the volume of the stone?
_____ cm³

No.5

A 4 by 4 by 4 cube is built from 64 small cubes. The outside of the big cube is painted yellow. Then it is taken apart.



Start here: every corner cube has 3 painted faces. Along one edge: 2 corners and _____ cubes between. On one face, the middle cubes make a _____ by _____ square. Fill in the table.

How many small cubes have paint on exactly ...

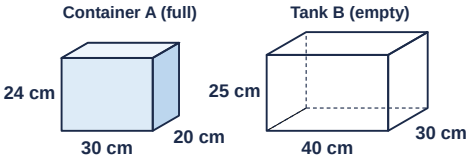
Painted faces	3 faces	2 faces	1 face
Number of cubes	8 corners	12 edges × _____ = _____	6 faces × _____ = _____

How many small cubes have no paint at all? .....

Check: do your four numbers add up to 64?

No.6

Container A is full of water. Tank B is empty.



a How many litres of water are in container A? _____ ℓ

b All of A is poured into B. How deep is the water in B? _____ cm

c Then some water is poured back from B into A until the water in A and B is equally deep. How deep is it?

.....

INVESTIGATION

Name _____ Group _____ Date _____

Investigation: Find all the ways

An open puzzle for everyone. Easy first step: list every box that holds exactly 12 cubes (for example 1 × 1 × 12). Then do part (a).

No.8

(a) A shop wants a box that holds exactly 48 small cubes of sugar, packed with no gaps. The box is a cuboid with whole-number edges in cubes. Find ALL the different boxes (a 2 by 4 by 6 box is the same as a 6 by 4 by 2 box). Make a list in order.

(b) Which box would you choose to sell, and why? Which is most like a cube?

(c) What if the shop packs 64 cubes? How many different boxes now? Is there a number of cubes that gives only ONE box shape?

My reflection (2 minutes. Your tutor may read this to plan the next lesson. It is never marked or shown to others. "Pass" is always OK.)

Something I did well

_____

The hardest part was

_____

What helped me: ☐ a picture or model ☐ a list or table ☐ a simpler number ☐ my partner ☐ I asked for help ☐ checking my estimate

Next time I will try

_____

☐ I am still stuck (we will look at it together next lesson, in class) ☐ Pass

Meet a mathematician: Archimedes (about 287 to about 212 BC, Syracuse, Sicily (ancient Greek world). Optional.)

Archimedes lived in Syracuse, Sicily, about 2300 years ago. He wrote a real book called On Floating Bodies and studied the volumes of solid shapes. A famous bath-and-wreath tale is told about him, but it was written down about 200 years later and historians doubt the details.

Try it (optional): A tank has a base 20 cm by 10 cm. A stone is dropped in and the water level rises by 2 cm. What is the volume of the stone?

My answer _____

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know. Draw a big, neat diagram first and label every length or angle you know.

Think harder

No.1

How many cubes with 2-cm edges can be packed in neat layers into a box 9 cm long, 6 cm wide and 5 cm high?

No.2

A tank is 1/3 full of water. After 12 ℓ more is poured in, it is 3/5 full. How many litres does the tank hold when full?

No.3

A cuboid has a square base and a height of 5 cm. Its volume is 245 cm^3. How long is each side of the base?

No.4

A cuboid 3 cm by 4 cm by 5 cm is built from 1-cm cubes and painted on the outside. How many cubes have no paint at all?

Go further (a puzzle to chew on)

No.5

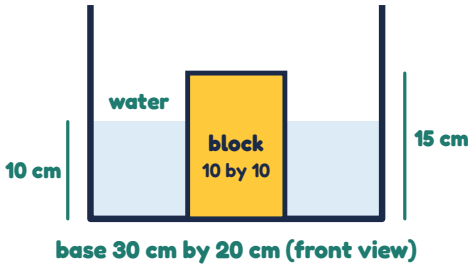
Three faces of a cuboid have areas 12 cm², 15 cm² and 20 cm². What is the volume of the cuboid?

No.6

A cuboid 3 cm by 4 cm by 5 cm is built from 1-cm cubes and painted on the outside. How many cubes have exactly 2 painted faces? How many have exactly 1?

No.7

A tank has a base 30 cm by 20 cm and water 10 cm deep. A block 10 cm by 10 cm by 15 cm is stood upright on the bottom. How deep is the water now?



base 30 cm by 20 cm (front view)

RATE RACERS

Name _____ Group _____ Date _____

Start here

After question 3: carry on, or try Think harder. Swap any time.

No.1

A machine fills 48 bottles every 3 minutes. (a) Complete the rate table.

Minutes	1	3	5		12
Bottles		48		160	

b The machine starts at 9.00 a.m. At what time has it filled 400 bottles? _____

No.2

Three shops sell the same can of drink.

SHOP A	SHOP B	SHOP C
Pack of 6 cans $4.20	Pack of 4 cans $2.92	Pack of 10 cans $7.80 + 2 cans FREE!

Start here: Shop A: $4.20 ÷ 6 = _____¢ a can. Shop B: $2.92 = 292¢, and 292 ÷ 4 = _____¢ a can.

a Price of 1 can: A _____ B _____ C _____

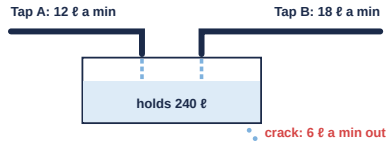
b Which shop is the best buy? _____

c Ben needs exactly 20 cans. He may mix shops, but only whole packs. Find one way and work out the cost.

Start here: take Shop C's 12 cans for $7.80. How many more cans does Ben need? Which pack makes them?

No.3

An empty tank holds 240 ℓ.



(a) Only tap A is turned on. How long does it take to fill the tank?

_____

(b) Both taps are turned on. How long does it take now?

_____

(c) *(try this one if you like)* Both taps are on and the crack lets out 6 ℓ a minute. How long to fill the tank? _____

(d) *(try this one if you like)* The tank is full. The taps are off but the crack still leaks. How long until it is half full? _____

Start here: page 2

Page 2. Start with Q4 (a), Q6 (a) and then Q5, then choose. You do not have to finish. Last 10 minutes: Think harder, Go further or the Investigation.

No.4

Printer P prints 30 pages a minute. Printer Q prints 20 pages a minute. A job has 900 pages.

a Both printers start together. How long do they take to print the job? _____

b Instead, P starts alone. After 5 minutes, Q joins in. How long does the whole job take from the start?

c Mei splits the 900 pages so that P and Q start together and finish together. How many pages does P get?

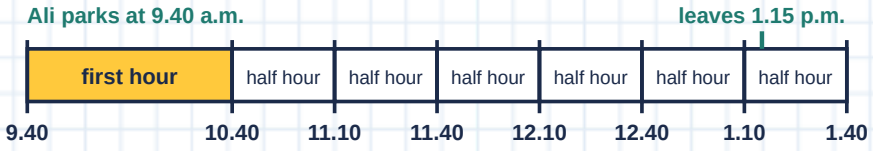
_____

No.5

Look at the car park sign.



Start here: circle every half hour Ali's car is in the car park (a part of a half hour counts), then count them.



a Ali parks from 9.40 a.m. to 1.15 p.m. How much does he pay?

b Mei pays $3.60. What is the longest time she could have parked? _____

No.6

Gym A charges $30 to join and then $4 for each visit. Gym B has no joining fee and charges $7 for each visit.

a Complete the table.

Number of visits	5	10	15
Gym A costs			
Gym B costs			

b For how many visits do the two gyms cost the same? _____

c Siti plans to go 14 times. Which gym is cheaper for her, and by how much?

_____

INVESTIGATION

Name _____ Group _____ Date _____

Investigation: Find all the ways

An open puzzle for everyone. Easy first step: tap A runs for 6 minutes. How many litres go in? How many litres are still needed, and how many minutes of tap B give that? Then do part (a).

No.8

(a) Tap A puts 2 litres a minute into an empty tank. Tap B puts 3 litres a minute. The taps are used ONE AT A TIME, each for a whole number of minutes (0 is allowed). Find ALL the ways to put exactly 24 litres into the tank. Make a table.

(b) Which way takes the shortest total time? Why does the faster tap win?

(c) What if the tank needs 25 litres? How many ways now? What changes and what stays the same?

My reflection (2 minutes. Your tutor may read this to plan the next lesson. It is never marked or shown to others. "Pass" is always OK.)

Something I did well

_____

The hardest part was

_____

What helped me: ☐ a picture or model ☐ a list or table ☐ a simpler number ☐ my partner ☐ I asked for help ☐ checking my estimate

Next time I will try

_____

☐ I am still stuck (we will look at it together next lesson, in class) ☐ Pass

Meet a mathematician: Mary Jackson (1921–2005, United States. Optional.)

Mary Jackson was born in 1921 in Hampton, Virginia. She began as a human computer at the NACA laboratory at Langley, then worked on a wind tunnel for model aircraft. After graduate-level courses she became an aerospace engineer in 1958.

Try it (optional): A model plane flies 840 km in 3 hours at a steady rate. How far does it fly in 1 hour? At that rate, how far would it fly in 5 hours?

My answer _____

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Think harder

No.1

A tap fills a tank in 12 minutes at 15 ℓ a minute. When a second tap is also turned on, the tank fills in 9 minutes. How many litres a minute does the second tap fill?

No.2

Eggs are sold in boxes of 10 for $3.30 and boxes of 6 for $2.10. Siti needs at least 28 eggs. What is the least she can pay?

No.3

A car park charges $1.50 for the first hour and $1 for every half hour or part of a half hour after that. How much does it cost to park from 10.15 a.m. to 2.50 p.m.?

No.4

A train travels 150 km in 2 hours 30 minutes and does not change its speed. How far does it travel in 1 hour 45 minutes?

Go further (a puzzle to chew on)

No.5

3 cats eat 3 fish in 3 minutes. At this rate, how many cats are needed to eat 100 fish in 100 minutes?

No.6

Candle A is 20 cm tall and burns down 2 cm every hour. Candle B is 14 cm tall and burns down 1 cm every hour. They are lit at the same time. After how many hours are they the same height? How tall are they then?

No.7

A tap fills an empty bath in 12 minutes. With the plug out, a full bath empties in 20 minutes. Ben turns on the tap but forgets the plug. How long does the bath take to fill?

PERCENTAGE PUZZLES

Name _____ Group _____ Date _____

Start here

After question 3: carry on, or try Think harder. Swap any time.

No.1

Percent means "out of 100". Build percentages of $240 from 10%, 5% and 1%. Fill in the table.

Percentage	10%	5%	1%	15%	35%	99%
of $240						

a How did you find 35% from the boxes you already had? _____

b A quick way to find 99% of $240: _____

No.2

Mei wants a rice cooker. Two shops sell the same one. GST is 9%.

Shop A
$160 before GST
20% OFF today!
9% GST added at the till

Shop B
$140
GST already included
No sale today

Start here: find 10% of Shop A's price by dividing by 10. Then 20% is double that.

a Shop A: the price after 20% off, before GST. _____

b Shop A: the GST on the sale price. _____

c Shop A: the price Mei pays. _____

d Which shop is cheaper? By how much? _____

No.3

A T-shirt cost $80. The shop raised its price by 25%. A month later, it cut the new price by 20%.

a The price after the 25% rise: _____ (b) The price after the 20% cut: _____

c Siti says, "Up 25% and then down 20% is the same as up 5%, so the final price is $84." Is she right? Explain.

.....

Start here: page 2

Page 2. Start with Q4 (a) to (c) and Q5, then choose. You do not have to finish. Last 10 minutes: Think harder, Go further or the Investigation.

No.4

Siti had $120. She spent 20% of her money on a bag. Then she spent 25% of the remaining money on a book.

Start here: 10% of $120 is $_____, so 20% of $120 is $_____.



Each part is 10% of $120. Shade the bag. Then split what is left into 4 equal parts.

a The bag cost _____ (b) The book cost _____ (c) Money left _____

d Ben says, "She spent 20% + 25% = 45%, so 55% of her money is left." What percentage of her money is really left? Explain.

No.5

There are 40 pupils in a class. 60% of them are girls. 25% of the girls and 50% of the boys wear glasses. Fill in the table.

	Girls	Boys	Whole class
Number of pupils			40
Wear glasses			

a What percentage of the whole class wear glasses? _____

INVESTIGATION

Name _____ Group _____ Date _____

Investigation: Is it always true? Test and explain

An open puzzle for everyone. Easy first step: on a $100 bag take 20% off (that is $20) and write the new price. Then take 10% off that new price. Then do part (a) the other way round.

No.8

(a) A $100 bag is on sale. Shop 1 takes 20% off and then a further 10% off the new price. Shop 2 takes 10% off first and then a further 20% off the new price. Work out both prices. Is one cheaper?

(b) Test two more pairs of discounts of your own (for example 50% and 10%, or 30% and 20%). Does the ORDER of two discounts ever change the final price? Is it always true?

(c) Is 20% off then 10% off the same as 30% off? Test and explain. (Hint: after 20% off, you pay 4/5. After 10% off, you pay 9/10.)

My reflection (2 minutes. Your tutor may read this to plan the next lesson. It is never marked or shown to others. "Pass" is always OK.)

Something I did well

_____

The hardest part was

_____

What helped me: ☐ a picture or model ☐ a list or table ☐ a simpler number ☐ my partner ☐ I asked for help ☐ checking my estimate

Next time I will try

_____

☐ I am still stuck (we will look at it together next lesson, in class) ☐ Pass

Meet a mathematician: Adam Ries (1492–1559, Germany. Optional.)

Adam Ries was born in what is now Germany around 1492 and ran a calculation school in Erfurt. He wrote arithmetic books in German, not Latin, so apprentices could read them, with everyday examples about prices, wages, profit and loss, and interest. In Germany people still say a sum is right "nach Adam Riese".

Try it (optional): A bag costs $200. A shop takes off 20% as a discount. What is the sale price? Then 9% GST is added to the sale price. What is the final price?

My answer _____

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Think harder (Q1: preview of P6, percentage change)

No.1

At a hawker centre, a plate of chicken rice went up from $4 to $5. A month later, it went back down from $5 to $4. (a) Find the percentage increase from $4 to $5. (b) Find the percentage decrease from $5 to $4. (c) Both changes are $1. Why are the two percentages different? Hint: percentage change = the change out of the amount you START with.

No.2

A box of cereal costs $15 in two shops. Shop X: "Buy 3, get 1 free!" Shop Y: "30% off every box!" Mei buys 4 boxes. (a) How much does she pay at each shop? Which is the better buy, and by how much? (b) Shop X's deal is really how many percent off?

No.3

A $60 shirt gets 20% off and a $40 hat gets 10% off. Then 9% GST is added to the total of the two sale prices. How much does Ali pay?

No.4

Siti puts $2000 in a bank that pays 3% simple interest each year, always on the $2000 she started with. After how many years is the total interest first MORE than $200? How much interest is that?

Go further (a puzzle to chew on: P6 preview)

No.5

A price tag says $109, GST included. GST is 9%. Ali says the GST is 9% of $109, which is $9.81. Is he right? How much GST is in the price?

No.6

An $80 lamp is on sale in two shops. Shop P takes 25% off. Shop Q takes 10% off, and then a further 15% off the new price. In which shop is the lamp cheaper, and by how much?

No.7

Mrs Tan spent 40% of her money on groceries and 25% of the remaining money on a gift. She had $54 left. How much money did she have at first?

ANGLE CHASERS

Name _____ Group _____ Date _____

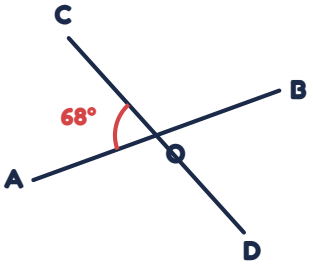
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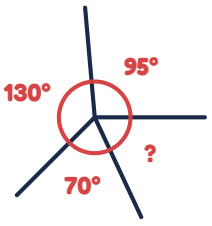
After question 3: carry on, or try Think harder. Swap any time.

Reading angles: angle BOC is the angle at O between OB and OC (the middle letter is the corner).
Reasons: A straight line 180°. B point 360°. C vertically opposite angles are equal. D triangle 180°. E isosceles: base angles are equal. F parallelogram or trapezium: neighbours add to 180°.

No.1

Left: straight lines AB and CD cross at O. Right: four angles meet at a point. Drawings are not to scale, so do not use a protractor.





Hint: a straight line is a half-turn, 180°. A full turn is 360°. Where two lines cross, the angles opposite each other are equal (tear and check).

a Angle BOC = _____ (b) Angle BOD = _____ (c) Angle AOD = _____

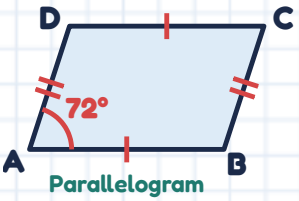
d The missing angle at the point on the right = _____. Which angle fact did you use?

_____

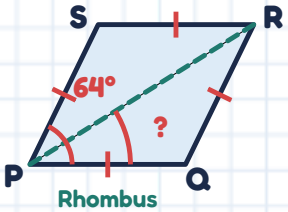
No.2

Find the missing angles. Write each angle fact you use.

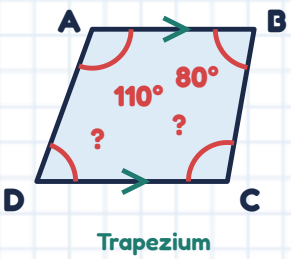
Hint: parallelogram: neighbours add to 180°, opposite angles are equal. Rhombus: 4 equal sides, so each diagonal makes isosceles triangles. Trapezium: angles between the parallel sides add to 180°. **Write like this:** angle B = 108° because neighbours in a parallelogram add to 180°.



Parallelogram



Rhombus



Trapezium

a Parallelogram: angle B = _____ angle C = _____ angle D = _____

b (try this one if you like) Rhombus: angle PQR = _____ angle QPR = _____ (hint: PQ = QR)

c Trapezium: angle D = _____ angle C = _____

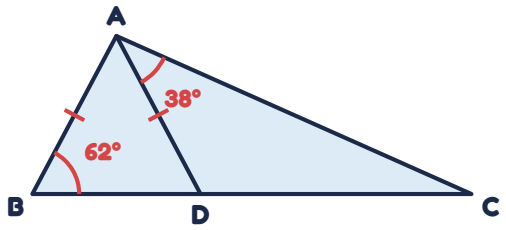
Start here: page 2

Page 2. Start with Q3 (a) (b) and Q4 (a) (b), then choose. You do not have to finish. Last 10 minutes: Think harder, Go further or the Investigation.

Reasons: A straight line 180°. B point 360°. C vertically opposite angles are equal. D triangle 180°. E isosceles: base angles are equal. F parallelogram or trapezium: neighbours add to 180°.

No.3

In triangle ABC, D is on BC and AB = AD. Angle B = 62° and angle DAC = 38°. Find each angle. Give a reason (a letter A to F, or words) for (b) and (d) only.



a Angle ADB = _____

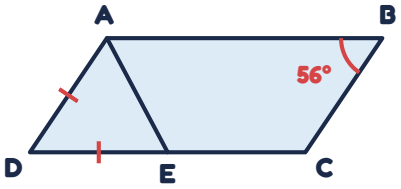
b Angle BAD = _____ because _____

c Angle ADC = _____

d Angle ACD = _____ because _____

No.4

ABCD is a parallelogram. E is a point on DC with DA = DE. Angle ABC = 56°.



a ADC = _____ (b) DEA = _____ (c) AEC = _____ (d) EAB = _____

Show how you found angle EAB: _____

No.5

An isosceles triangle has one angle of 50°. What could the other two angles be? Find every case.

a List both: _____ and _____

b An isosceles triangle has one angle of 100°. How many different triangles can you make? Explain.

.....

INVESTIGATION

Name _____ Group _____ Date _____

Investigation: Find all the ways

An open puzzle for everyone. Easy first step: start with a base angle of 10°. The other base angle is also 10°. What is the top angle? Write it as three angles. Then try 20°.

No.8

(a) An isosceles triangle has all three angles equal to a whole number of TENS of degrees (10°, 20°, 30° and so on). Find ALL the possible triangles. Write each as three angles.

(b) How many of your triangles have an angle bigger than 90°? Which one is equilateral?

(c) What if the angles are multiples of 15° instead? How many triangles now? What stays the same in your method?

My reflection (2 minutes. Your tutor may read this to plan the next lesson. It is never marked or shown to others. "Pass" is always OK.)

Something I did well

_____

The hardest part was

_____

What helped me: ☐ a picture or model ☐ a list or table ☐ a simpler number ☐ my partner ☐ I asked for help ☐ checking my estimate

Next time I will try

_____

☐ I am still stuck (we will look at it together next lesson, in class) ☐ Pass

Meet a mathematician: Maryam Mirzakhani (1977–2017, Iran and the United States. Optional.)

Maryam Mirzakhani grew up in Tehran, Iran, and won gold medals at the International Mathematical Olympiad in 1994 and 1995. She often thought by drawing big sketches, and she studied the paths of a ball bouncing around a billiard table. In 2014 she became the first woman to win the Fields Medal.

Try it (optional): A ball hits a straight wall. Its path makes a 35° angle with the wall on the way in, and the same 35° on the way out. What is the angle between the path going in and the path going out?

My answer _____

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know. Draw a big, neat diagram first and label every length or angle you know.

Think harder

No.1

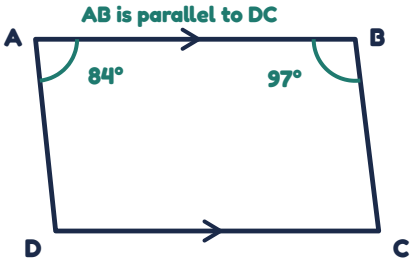
In parallelogram ABCD, angle A is 3 times angle B. Find angle A and angle B. Give your reason.

No.2

An isosceles triangle has an angle of 110°. Find the other two angles. Why is there only one answer?

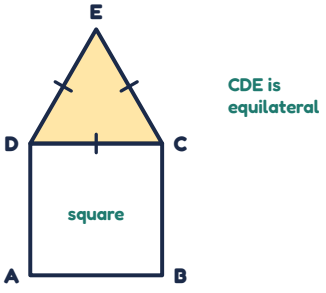
No.3

In trapezium ABCD, AB is parallel to DC. Angle DAB is 84° and angle ABC is 97°. Find angle ADC and angle BCD.



No.4

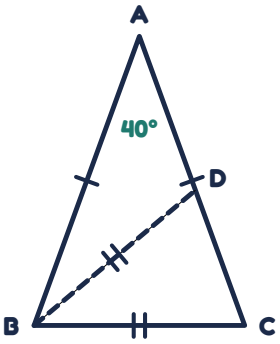
ABCD is a square. CDE is an equilateral triangle, and E is outside the square. Find angle ADE and angle DAE.



Go further (a puzzle to chew on)

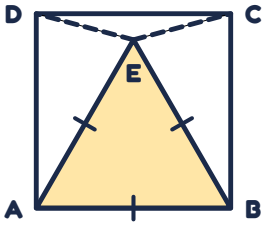
No.5

ABC is an isosceles triangle with AB = AC and angle BAC = 40°. D is a point on AC with BD = BC. Find angle ABD.



No.6

ABCD is a square. E is a point inside it so that ABE is an equilateral triangle. Find angle DEC.



No.7

In an isosceles triangle, one angle is 30° bigger than another angle. Find all the possible sets of three angles.

PATTERN GENERALISERS

Name _____ Group _____ Date _____

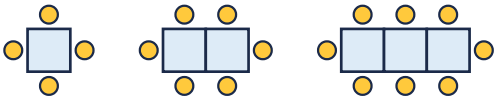
Start here

After question 3: carry on, or try Think harder. Swap any time.

A growing pattern: find what stays the same and what grows each time. Then say the rule in words, like this: number of sticks = 3 × figure number + 1.

No.1

At a hawker centre, square tables are pushed together in a row. One chair goes on each free side.



a 4 tables: _____ chairs. 5 tables: _____ chairs. 10 tables: _____ chairs.

b Rule in words: chairs = _____ × tables + _____

c How many tables are needed to seat exactly 50 people? _____

d Ben says, "10 tables seat 22 people, so 20 tables seat 44." Is he right? Explain.

.....

No.2

Matchstick houses are joined in a row.



a Each new house adds _____ sticks. Why not 6? _____

b Rule: sticks = _____ × houses + _____. For 20 houses: _____ sticks.

c Mei has 200 sticks. Greatest number of houses she can make? Sticks left over? _____

Start here: page 2

Page 2. Start with Q3 and Q4 (a) (b), then choose. You do not have to finish. Last 10 minutes: Think harder, Go further or the Investigation.

No.3

Guess and check with a table. Adult tickets to the Science Centre cost $12 each and child tickets cost $7 each. A group buys 15 tickets for $145 altogether. How many adult tickets do they buy?

Adult tickets	Child tickets	Total cost	Too much or too little?
5	10	$60 + $70 = $130	too little

Answer: _____ adult tickets and _____ child tickets.

No.4

Siti has 30 coins. They are all 20¢ and 50¢ coins, worth $10.20 altogether.

a If all 30 coins were 20¢: worth _____. Each change of a 20¢ for a 50¢ adds _____.

b How many 50¢ coins does Siti have? Show your working and check.

No.5

A quiz has 20 questions. Each correct answer scores 5 marks. For each wrong answer, 2 marks are taken away. Ali answered every question and scored 65 marks.

a How many questions did Ali get correct? Hint: suppose he got them all correct first.

b Mei also answered every question and says she scored 80 marks. Why must she be wrong? Start here: all correct scores 100. Each wrong answer takes off 7. Write the scores for 0, 1, 2, 3 wrong: 100, _____, _____, _____, _____. Is 80 in your list?

INVESTIGATION

Name _____ Group _____ Date _____

Investigation: Is it always true? Test and explain

An open puzzle for everyone. Easy first step: fill in the first three totals: 1 = 1, 1 + 3 = 4, 1 + 3 + 5 = ____. Then carry on.

No.8

(a) Add the first odd numbers: 1, then 1 + 3, then 1 + 3 + 5, then 1 + 3 + 5 + 7. Make a table of the first 6 totals. What do you notice?

(b) Is it always true? Use a picture of square dots (add an L-shaped border each time) to explain why.

(c) What if you add the first EVEN numbers instead (2, then 2 + 4, then 2 + 4 + 6 ...)? Find a rule in words. Use it to predict the total of the first 20 even numbers, then check it another way.

My reflection (2 minutes. Your tutor may read this to plan the next lesson. It is never marked or shown to others. "Pass" is always OK.)

Something I did well

_____

The hardest part was

_____

What helped me: ☐ a picture or model ☐ a list or table ☐ a simpler number ☐ my partner ☐ I asked for help ☐ checking my estimate

Next time I will try

_____

☐ I am still stuck (we will look at it together next lesson, in class) ☐ Pass

Meet a mathematician: Al-Karaji (about 953–1029, Baghdad (today Iraq). Optional.)

Al-Karaji lived about a thousand years ago and spent most of his life in Baghdad. In his algebra book Al-Fakhri he explained rules for working with unknown numbers. He also wrote about a pattern: add the cubes of 1, 2, 3 and so on, and you get the square of the sum of those numbers.

Try it (optional): Test Al-Karaji's pattern. Work out 1 × 1 × 1 + 2 × 2 × 2 + 3 × 3 × 3 + 4 × 4 × 4. Then add 1 + 2 + 3 + 4 and multiply the answer by itself. What do you notice? Predict the answer for up to 5.

My answer _____

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Think harder

No.1

Beads repeat in the order red, red, blue, yellow, yellow, yellow. What colour is the 50th bead? How many yellow beads are in the first 100?

No.2

40 square tables in one long row seat 2 × 40 + 2 = 82 people. How many can sit if the tables are put in 8 rows of 5 tables instead?

No.3

A farm has chickens and goats. There are 35 heads and 94 legs. How many goats are there?

No.4

The pattern 5, 8, 11, 14, ... keeps going up by 3. Is 101 in the pattern? Is 100? Explain how you know, and say which number in the pattern it is if it is there.

Go further (a puzzle to chew on)

No.5

A dot pattern makes diamonds. Figure 1 has 1 dot. Figure 2 has rows of 1, 3 and 1 dots (5 dots). Figure 3 has rows of 1, 3, 5, 3 and 1 dots (13 dots). How many dots are in figure 6?

No.6

A quiz has 25 questions. A correct answer scores 4 marks, 1 mark is taken away for each wrong answer, and a blank scores 0. Mei left 3 questions blank and scored 68. How many did she get correct?

No.7

Ali writes the counting numbers one after another without spaces: 123456789101112131415... What is the 100th digit he writes?

CHALLENGE DAY

Name _____ Group _____ Date _____

Start here

After question 3: carry on, or try Think harder. Swap any time.

Choose a start: draw a model, list in order, make a table, work backwards or guess and check. Check EVERY clue.

No.1

Logic grid. Ali, Ben, Mei and Siti each ordered a different dish at a hawker centre. Use the clues to find who ordered what. Put a tick for yes and a cross for no.

Clue 1: Ali did not order laksa or satay.
Clue 2: The roti prata was ordered by a girl (Mei and Siti are girls; Ali and Ben are boys).
Clue 3: Ben did not order satay. Clue 4: Mei did not order roti prata.

	Chicken rice	Laksa	Satay	Roti prata
Ali				
Ben				
Mei				
Siti				

Ali: _____ Ben: _____ Mei: _____ Siti: _____

No.2

Working backwards. Ali spent 1/3 of his money on a book. He then spent $12 on lunch. He spent half of what was left on a gift. Ali had $15 left at the end. How much money did he start with?

Start here: write the steps in order. Then undo the LAST step first.

Answer: $ _____

No.3

Draw a model. Mei and Ravi have $115 altogether. Ravi has $10 more than 3/4 of Mei's money. How much money does each have? Check your answer against BOTH clues.

Start here: Mei is 4 equal units. How many units is 3/4 of Mei's? Where does the $10 go?

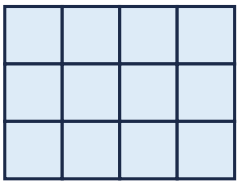
Mei: $_____ **Ravi: $**_____ _____

Start here: page 2

Page 2. Start with Q4, Q5 and Q6 (a), then choose. You do not have to finish. Last 10 minutes: Think harder, Go further or the Investigation.

No.4

How many squares of ALL sizes can you find in this 3 by 4 grid? Count by size so you do not miss any.



3 rows of 4 small squares

a Squares of size 1 by 1: _____

b Squares of size 2 by 2: _____ Squares of size 3 by 3: _____

c How many squares altogether? How do you know there are none bigger?

.....

No.5

Ali, Ben and Mei were playing. One of them broke a vase. Ali says, "Ben broke it." Ben says, "Mei broke it." Mei says, "I did not break it." Exactly ONE of the three children is lying. Who broke the vase? Try each child and count the liars.

If the vase was broken by ...	Is Ali lying?	Is Ben lying?	Is Mei lying?
Ali			
Ben			
Mei			

Answer: _____ broke the vase.

No.6

Mum is 41 years old and Mei is 9 years old.

a In how many years will Mum be exactly 3 times as old as Mei? Show your table or working.

.....

.....

b Which start did you use? Why did you choose it?

.....

INVESTIGATION

Name _____ Group _____ Date _____

Investigation: Find the pattern

An open puzzle for everyone. Easy first step: with 1 step there is 1 way. With 2 steps there are 2 ways (1 + 1, or 2). Draw the ways for 3 steps first.

No.8

(a) Ali climbs a staircase and takes either 1 step or 2 steps at a time. How many different ways can he climb a staircase with 1 step? 2 steps? 3 steps? 4 steps? List every way for 4 steps (for example 1+1+2 and 2+1+1 are different).

(b) Make a table up to 6 steps. What pattern do you see? Explain why it works (hint: think about the LAST jump Ali makes).

(c) Use your pattern to predict the number of ways for 8 steps. What if Ali could also take 3 steps? Try 3 and 4 steps again.

My reflection (2 minutes. Your tutor may read this to plan the next lesson. It is never marked or shown to others. "Pass" is always OK.)

Something I did well

_____

The hardest part was

_____

What helped me: ☐ a picture or model ☐ a list or table ☐ a simpler number ☐ my partner ☐ I asked for help ☐ checking my estimate

Next time I will try

_____

☐ I am still stuck (my parent can ask my tutor about it) ☐ Pass

Meet a mathematician: George Polya (1887–1985, Hungary, Switzerland and the United States. Optional.)

George Polya was born in Budapest, Hungary, in 1887 and later taught in Switzerland and the United States. His 1945 book How to Solve It gave four steps for any problem: understand it, make a plan, carry it out, and look back at your work. His helpful ideas include drawing a picture and working backwards.

Try it (optional): Work backwards. I think of a number, double it and add 6. I get 20. What was my number?

My answer _____

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Think harder

No.1

Siti spent 2/5 of her money, then 1/3 of what was left. She had $24 left at the end. How much did she start with?

No.2

How many 3-digit numbers have digits that add up to 5? List them all in order.

No.3

The sum of two consecutive numbers (numbers that are next to each other, like 40 and 41) is 145. What are the two numbers?

No.4

A snail climbs 3 m up a 10 m wall each day, and slips down 2 m each night. On which day does it reach the top?

Go further (a puzzle to chew on)

No.5

A water tank is 1/3 full. After 20 litres are added it is 1/2 full. Then 1/4 of the water in the tank is let out. How many litres of water are in the tank now? What fraction of the tank is full?

No.6

25 coins of 50¢, 10¢ and 20¢ are worth $5.30 altogether. There are twice as many 10¢ coins as 50¢ coins. How many coins of each kind are there?

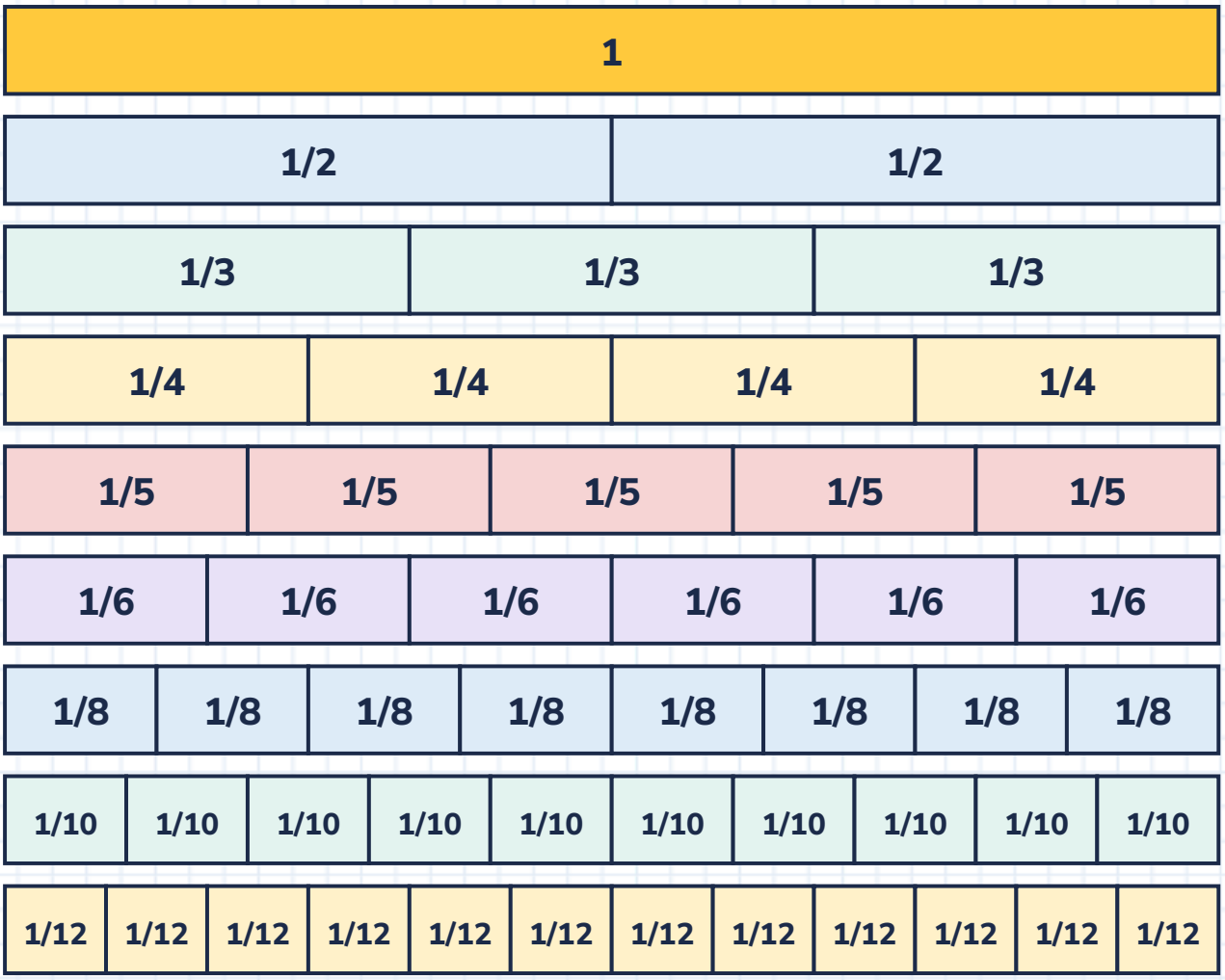
No.7

Ali, Ben, Mei and Siti sit in a row of 4 chairs. Mei is not on either end. Ben does not sit next to Mei. Siti sits somewhere to the left of Ali. Ben is not on the left end. In what order do they sit, from left to right?

Cut-outs

For the Level 5 program. Cut along the dashed lines and keep the pieces in one envelope.

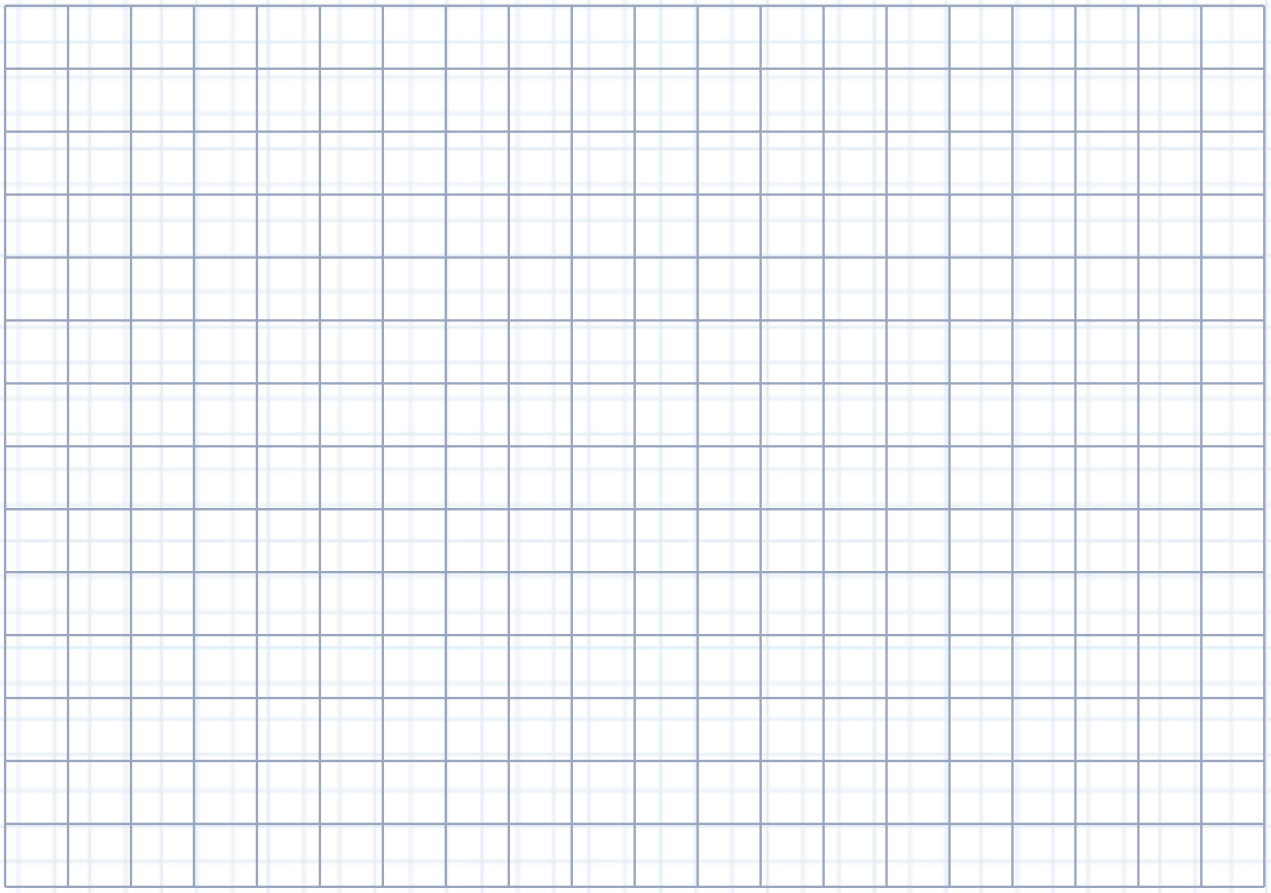
Fraction wall (Lessons 2 and 3)



Line the strips up under the whole. Which fractions are the same size?

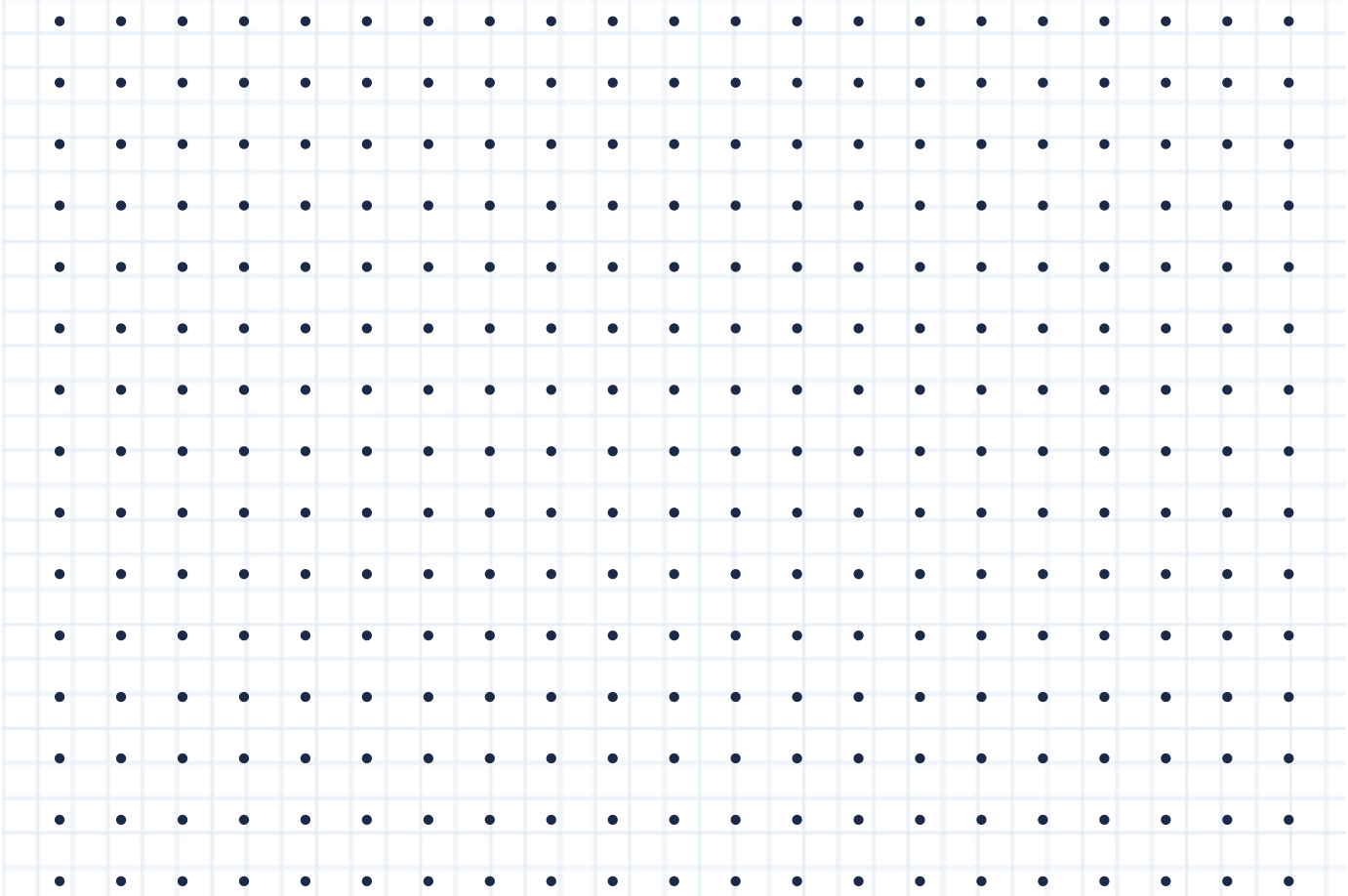
Square grid

For Lesson 4 (triangle areas).



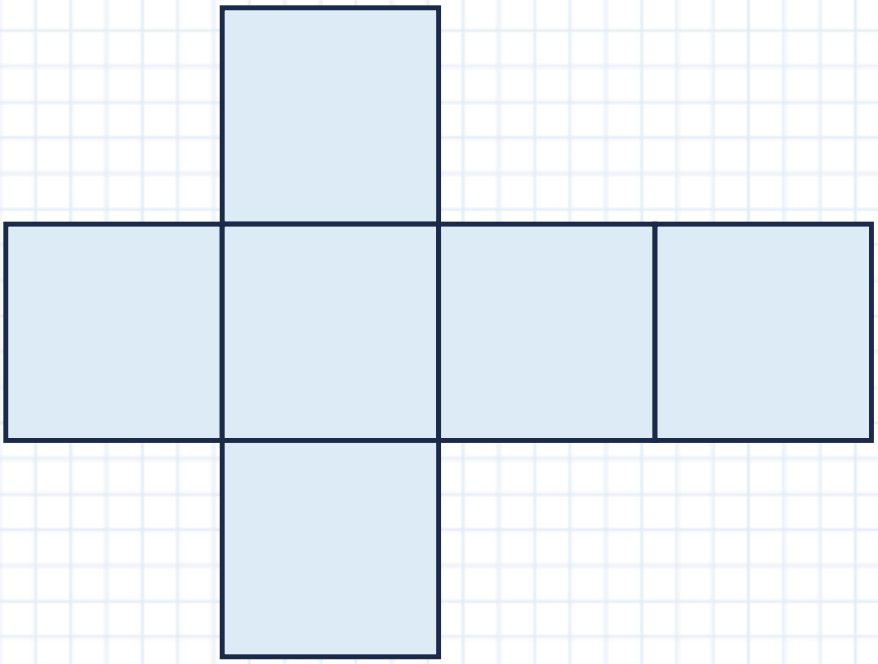
Dot grid

For Lesson 8.



Cube net

For Lesson 5 (volume).



Cut round the outside, fold on the lines and tape: a cube. Try drawing other nets on the square grid.

CERTIFICATE OF COMPLETION

Math Inquirers Holiday Program

Level 5 Holiday Program

This certifies that

.....

completed all 10 lessons, solved puzzles, spotted patterns and explained their thinking.

Notice it. Test it. Explain why.

Tutor: _____ Date: _____